Week 10 – Incremental vs Transformational Research, and Imagination (Lean Notes)

No transcript was available for this week, so these notes are built from the Week 10 assignment and its answer key only. They cover everything the assignment tests; lecture examples that the assignment does not mention may be missing. ⭐ = tested in the assignment (every card is).

1. Concept cards

1. What school expects ⭐

- Students in school expect **well-defined questions**, a **question-paper pattern fixed in advance** and **clarity on the "portions"** (syllabus). They do **not** expect questions from outside the syllabus.
- The exam format is useful: it helps you **understand fundamentals**, **break a problem into parts and solve them**, and **prepares you for a professional setting**. It does **not** teach you to handle unexpected questions, because the format avoids them.

2. A "poorly defined" problem ⭐

- Example: "How long will it take to go from Chennai to Kolkata?" The answer depends on things the question does not say: **mode of travel, route, speed and start time**. So it is **poorly defined** and has **no unique answer**; different assumptions give different answers.
- Recipe: if a question leaves out details, it is poorly defined; a school question would state them.

3. What the workplace and college need ⭐

- A professional workplace requires you to **handle open-ended questions** and **solve poorly defined problems**. A **syllabus** and a **question-paper pattern** belong to school, not work.
- College should **prepare you for professional life** and **teach you to handle open-ended problems**. It should not focus **only** on well-defined problems or just repeat the school approach.

4. Three kinds of research ⭐

Kind	Meaning	Example from the assignment
Incremental	extends existing knowledge by a **small step**	applying a known theory to **one additional new situation**
Systematic	a **planned, orderly series** of studies of one phenomenon	a series of experimental techniques on the **same phenomenon across different samples**
Transformational (breakthrough)	**changes what is possible** or how people use something	see card 5

5. Spotting transformational change ⭐

- **Phones:** the jump from **cordless phone to mobile phone** is transformational (it removed dependence on a fixed place). The first landline, the first cordless phone and making early mobiles **smaller or more colourful** are not (they are early steps or refinements).
- **Bicycles: adding gears** is transformational (it changes what the bicycle can do: climb slopes, cover distance). A cushioned or ergonomic seat or a carrier only improves comfort or convenience.
- Test: does it change **capability**, or only comfort, looks, size or convenience?

6. Imagination in research ⭐

- Imagination means **going beyond what nature or the current solution shows**. Airplane example: **designing fixed wings** although birds have flapping wings. Adding seats, a cargo version or more engines are practical refinements or extensions, not imagination.

2. Top traps (most likely first)

1. **Transformational means a change in capability** (cordless to mobile, gears), not comfort, size, colour or convenience.
2. **Incremental** = a small step, such as a known theory applied to one more case.
3. A **series of techniques on one phenomenon across samples** is **systematic**, not incremental (the answer key says so).
4. School expects defined questions and portions; **work and research are open-ended and poorly defined**.
5. The **exam format does not teach handling unexpected questions**, even though it builds fundamentals.
6. "Focus **only** on well-defined problems" is wrong for college (the word "only").
7. The portal cuts off Q2's option D after "Has a unique"; it is presumably "Has a unique answer", which is wrong.

3. Assignment questions with solutions

Q1. Students in schools typically expect (select all that apply) *(Q1)* - A. Well defined questions B. Question paper patterns to be finalized in advance C. Clarity on the "portions" for an exam D. Questions outside the syllabus

Solution (card 1): D is the one students do not expect. **Answer: A, B and C.**

Q2. From the perspective of a school student the question "How long will it take to go from Chennai to Kolkata" is *(Q2)* - A. A "poorly" defined question B. Is a well defined question C. Is typically how a full question is framed in an exam D. Has a unique (cut off by the portal)

Solution (card 2): mode, route, speed and start time are missing, so it is poorly defined. **Answer: A.**

Q3. The format of exams in schools (select all that apply) *(Q3)* - A. Helps you understand fundamentals B. Teaches you to breakdown a problem to parts and solve them C. Prepares you for a professional setting D. Teaches you to handle unexpected questions

Solution (card 1): D is wrong because the format avoids unexpected questions. **Answer: A, B and C.**

Q4. A typical professional workplace requires you to (select all that apply) *(Q4)* - A. Handle open ended questions B. Solve "poorly" defined problems C. Focus on syllabus D. Work within a question paper pattern

Solution (card 3): a syllabus and paper pattern belong to school. **Answer: A and B.**

Q5. Learning in college is expected to (select all that apply) *(Q5)* - A. Prepare you for professional life B. Focus only on well-defined problems C. Teach you to handle open ended problems D. Follow the same approach used in school

Solution (card 3): B has "only" and D repeats school. **Answer: A and C.**

Q6. Applying a known theory to one additional new situation is most likely *(Q6)* - A. Breakthrough research B. Incremental research C. Transformational research D. Important contribution to research

Solution (card 4): a small extension of existing knowledge is incremental. **Answer: B.**

Q7. Use of a series of experimental techniques to study the same phenomenon across different samples is typically an example of: *(Q7)* - A. Systematic research B. Transformational research C. Incremental research D. Breakthrough research

Solution (card 4): a planned, orderly programme on one phenomenon is systematic. **Answer: A.**

Q8. In the development of phones, which can be considered transformational? *(Q8)* - A. First deployment of the landline B. First deployment of the cordless phone C. The improvement from cordless phone to mobile phone D. Making the first-generation mobile phones smaller and more colorful

Solution (card 5): the mobile phone removed dependence on a fixed place; D only refines. **Answer: C.**

Q9. In the development of bicycles, which can be considered transformational? *(Q9)* - A. Improvement from wooden seat to cushioned seat B. Making the seat more ergonomic C. Adding a carrier D. Adding gears to a bicycle

Solution (card 5): gears change the capability; the others are comfort or convenience. **Answer: D.**

Q10. In the development of airplanes, we can associate imagination with *(Q10)* - A. Designing fixed wings even though birds have flapping wings B. Adding seats inside an airplane C. Creating a cargo version of the airplane D. Adding more engines to the airplane

Solution (card 6): going beyond nature's design is imagination; B, C and D are refinements. **Answer: A.**